SHREERAJ PAWAR

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EDUCATION

Bachelor of Science, The University of Alabama, Tuscaloosa, AL August 2026
Major: Mechanical Engineering, Minors: Manufacturing, Mathematics — GPA: 4.00/4.00

SKILLS

Reliability & Continuous Improvement: Lean Six Sigma Green Belt, Root-Cause Troubleshooting, Process-Improvement Assessment, Facility Systems Assessment (Compressed Air, Boiler, Motor)

Design & Manufacturing: Design for Manufacturability (DFM), BOM Management, Manufacturing Release, Cross-Functional Coordination

CAD & Analysis: SolidWorks, AutoCAD, Fusion 360, Onshape, ANSYS, Abaqus, HydroSym; CATIA v5 (in progress)

Programming: C, C++, Python, MATLAB, Microsoft Office

EXPERIENCE

Mechanical Engineering Intern, *The Boring Company*, Bastrop, TX Aug 2025 – May 2026

- Communicated effectively across cross-functional field and manufacturing teams to resolve manufacturability, structural performance, serviceability, and installation issues on heavy mechanical systems during active tunneling operations.
- Engineered and released mechanical components and support structures for Rock TBM systems — sensor mounts, operator platforms, protective shrouds, and structural hardware — using SolidWorks, producing full detail drawings, 2D sections, and parts documentation for manufacturing release, with hands-on involvement in production and testing.
- Created and managed large top-level TBM assemblies, detailed manufacturing drawings, and bills of materials, generating 2D and 3D CAD in SolidWorks from existing field hardware to support integration, fabrication, and installation while coordinating several parts and engineering changes concurrently.
- Designed a custom in-house TBM gripper system rated for extreme structural loads, applying first-principles hand calculations, FEA correlation, material selection, and formal design reviews to replace an outsourced solution.
- Developed vibration-conscious mounting and routing solutions for navigation systems, sensors, cables, and hoses under tight packaging constraints in harsh underground tunneling environments — direct DFM/design-for-assembly problem solving.

Design and Fabrication Lead, *Senior Design Team, University of Alabama*, Tuscaloosa, AL Aug 2024 – May 2025

- Led design, additive manufacturing, and physical testing (quasistatic and dynamic compressive loading) of tensegrity-like unit cells and lattices to optimize specific energy absorption through delocalization of forces.
- Directed simulation work in two CAD and analysis platforms (Abaqus and SolidWorks) to correlate theoretical and experimental results, supporting design iteration across a five-person team, and was responsible for weekly communication with faculty advisors.

Pollution Prevention Program Intern, *UA SafeState*, Tuscaloosa, AL May 2024 – Aug 2024

- Conducted process-improvement assessments of manufacturing operations at 3 Alabama facilities, identifying cost-saving and efficiency opportunities across production processes.
- Researched and presented process-improvement approaches biweekly to a 10–12 person team, cutting assessment turnaround time by 5% and raising resource-use efficiency by 15%.

Undergraduate Research Assistant, *Alabama Industrial Assessment Center*, Tuscaloosa, AL Apr 2023 – May 2025

- Guided a team of 4–6 students on industrial assessment visits, generating recommendations projected to save 2 million kWh of energy and $220,000 in costs, and reduce CO2 emissions by 1,316 tons annually.
- Authored detailed efficiency reports (ROI under 5 years) covering compressed air, boiler, motor, HVAC, and lighting systems; communicated findings directly to plant clients and implemented customer feedback and requests into further analysis.

PROJECTS

Design Team, *Alabama Rocketry Association*, Tuscaloosa, AL Aug 2021 – Apr 2022

- Designed and retrofitted structural components (bulkheads, inner body tube, centering rings, fins, thrust plate) for subscale and full-scale rockets, contributing to five milestone design review documents (PDR, CDR, FRR, LRR, PLAR).
- Performed carbon fiber layups on rocket nose cones and fins.

CERTIFICATIONS

Lean Six Sigma Green Belt
CATIA v5 Fundamentals — in progress